

Please make sure that you show all of the steps, not just the answer.

1) Find the first derivative $\frac{dy}{dx}$.

a) $y = e^x \sin x$

Product Rule
 $A'B + AB'$

$$A = e^x \quad A' = e^x$$

$$B = \sin x \quad B' = \cos x$$

$$(e^x)(\sin x) + (e^x)(\cos x)$$

$$\frac{dy}{dx} = e^x (\sin x + \cos x)$$

b) $y = \tan(x^2)$ Chain Rule

$$f(u) = \tan(u) \quad f'(u) \cdot u'$$

$$u = x^2$$

$$\frac{d}{dx} \tan(u) = \sec^2(u)$$

$$= \sec^2(x^2)$$

$$\frac{d}{dx} x^2 = 2x$$

$$\frac{dy}{dx} = 2x \sec^2(x^2)$$

c) $y = \ln(\ln x)^2$ Chain Rule

$$v = \ln(\ln x)$$

$$(v)^2$$

$$2v \cdot v'$$

$$2(\ln(\ln x)) \cdot [\ln(\ln x)]'$$

← chain Rule

$$\ln'(k) \cdot k'$$

$$k = \ln x$$

$$\left[\frac{1}{\ln x} \cdot \frac{1}{x} \right]$$

$$y' = 2(\ln(\ln x)) \cdot \frac{1}{\ln x} \cdot \frac{1}{x}$$

$$y' = \frac{2(\ln(\ln x))}{x \ln x}$$

d) $y = x\sqrt{e^x}$ Product Rule

$$A = x$$

$$B = \sqrt{e^x}$$

$$A' = 1$$

$$B' = \frac{1}{2} \sqrt{e^x}$$

$$A'B + AB'$$

$$y' = (1)(\sqrt{e^x}) + (x)(\frac{1}{2}\sqrt{e^x})$$

$$y' = \sqrt{e^x} + \frac{x}{2}\sqrt{e^x}$$

$$y' = \sqrt{e^x} \left(1 + \frac{x}{2} \right)$$

e) $y = \sqrt{\tan x}$ Chain Rule

$$y' = \sqrt{v} \cdot v'$$

$$= \frac{1}{2\sqrt{v}} \cdot v'$$

$$v = \tan x, \quad v' = \sec^2 x$$

$$y' = \frac{1}{2\sqrt{\tan x}} \cdot \sec^2 x$$

$$y' = \frac{\sec^2 x}{2\sqrt{\tan x}}$$

f) $y = \sqrt{x^2 + 2}$ Chain Rule

$$y' = (\sqrt{v})' \cdot v'$$

$$y' = \frac{1}{2\sqrt{v}} \cdot v'$$

$$v = x^2 + 2$$

$$y' = \frac{1}{2\sqrt{x^2 + 2}} \cdot (x^2 + 2)'$$

$$y' = \frac{1}{2\sqrt{x^2 + 2}} \cdot 2x$$

$$y' = \frac{2x}{2\sqrt{x^2 + 2}} \rightarrow$$

$$y' = \frac{x}{\sqrt{x^2 + 2}}$$

2) Find the smallest positive value for x so that $f'(x) = 0$.

a) $f(x) = \sin(\sqrt{x})$ Chain Rule

$$f(x)' = \sin'(u) \cdot u'$$

$$f(x)' = \cos(\sqrt{x}) \cdot \frac{1}{2\sqrt{x}}$$

$$f(x)' = \frac{\cos(\sqrt{x})}{2\sqrt{x}}$$

Now set it equal to zero:

$$2x \cdot \frac{\cos(\sqrt{x})}{2\sqrt{x}} = 0 \cdot 2x$$

$$\cos(\sqrt{x}) = 0 \quad \left. \begin{array}{l} \cos \text{ is } 0 \text{ at} \\ \sqrt{x} = \frac{\pi}{2} \dots \end{array} \right\}$$

$$\sqrt{x} = \frac{\pi}{2}$$

$$x = \left(\frac{\pi}{2}\right)^2$$

$$x = \frac{\pi^2}{4}$$

b) $f(x) = \sqrt{\sin x}$ Chain Rule

$$f(x)' = \frac{1}{2\sqrt{\sin x}} \cdot (\sin x)'$$

$$f(x)' = \frac{1}{2\sqrt{\sin x}} \cdot \cos x$$

$$f(x)' = \frac{\cos x}{2\sqrt{\sin x}} = 0$$

$$\cos x = 0$$

$$\rightarrow \text{at } \frac{\pi}{2} \dots$$

$$x = \frac{\pi}{2}$$

c) $f(x) = 5e^{-3x} + 7x$

$$f(x)' = \underbrace{5(e^{-3x})'}_{\text{Chain Rule}} + \underbrace{(7x)'}_{\text{Power Rule}}$$

$$(5)e^{-3x}(-3)$$

$$-15e^{-3x} + 7 = 0 \quad (\text{set to zero})$$

$$-15e^{-3x} = -7$$

$$e^{-3x} = \frac{7}{15}$$

$$-3x = \ln\left(\frac{7}{15}\right)$$

$$x = -\frac{1}{3} \ln\left(\frac{7}{15}\right)$$

d) $f(x) = \frac{x}{x^2+1}$

Quotient Rule

$$\frac{A'B - AB'}{B^2}$$

$$\begin{array}{cc} A & B \\ x & x^2+1 \\ A' & B' \\ 1 & 2x \end{array}$$

$$f(x)' = \frac{(1)(x^2+1) - (x)(2x)}{(x^2+1)^2}$$

$$f(x)' = \frac{x^2+1-2x^2}{(x^2+1)^2}$$

$$f(x)' = \frac{1-x^2}{(x^2+1)^2} \stackrel{(\text{set } = 0)}{=} 0 \quad \text{when the numerator is zero...}$$

$$0 = 1 - x^2$$

$$1 = x^2$$

$$\pm 1 = x \rightarrow \text{smallest}$$

$$x = 1$$

3) Find the equation of the line tangent to $f(x) = x^3$ when $x = 2$. Then use the equation of the tangent line to estimate 2.1^3 by substituting $x = 2.1$. How close is your estimate to the actual answer $2.1^3 = 9.261$?

$$f(x)' = 3x^2$$

at $x = 2$ the slope is

$$f(2)' = 3(2)^2 = 12$$

Now the point on the curve when $x = 2$

$$f(2) = (2)^3 = 8$$

- So the tangent line goes through $(2, 8)$
- with slope $m = 12$

Point Slope form:

$$y - 8 = 12(x - 2)$$

$$y = 12x - 16 \quad \text{tangent line}$$

$$y = 12(2.1) - 16 \quad \text{Estimate } (2.1^3) \text{ by plugging in } x = 2.1$$

$$y = 25.2 - 16$$

$$y = 9.2$$

$$\text{Estimate: } 9.2$$

$$\text{Actual Answer: } 9.261$$

$$\text{Difference: } 9.261 - 9.2 = 0.061$$